

ATHO

Monetary Supply Model

500-YEAR ISSUANCE AND FEE REFERENCE

Subsidy, tail emission, inflation, transaction fees, and miner revenue

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Pre-production notice: Ato is experimental software under active development. This paper describes the current repository; it is not a mainnet launch approval, investment statement, or claim of completed independent security assurance.

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Abstract

This document isolates Atho's monetary policy from the broader technical white paper and projects the current consensus subsidy schedule across 500 nominal years. It explains the monetary unit, reward eras, permanent tail emission, cumulative scheduled supply, annualized issuance, inflation calculation, coinbase maturity, transaction fee floor, optional tips, miner revenue, block capacity, hypothetical price conversion, and the difference between issued, circulating, burned, and economically available supply.

The projection uses the code-defined target cadence of one block every 100 seconds, or 315,360 expected reward-bearing blocks per nominal year. Under that cadence, year 500 ends at expected height 157,680,000 with 95,090,000 ATHO of scheduled subsidy issuance. Of that amount, 18,750,000 ATHO comes from the four bootstrap eras and 76,340,000 ATHO comes from permanent tail emission. The tail reward remains 0.50 ATHO per block and produces 157,680 ATHO of scheduled annual issuance. Relative to year-500 scheduled supply, that is approximately 0.165821853 percent.

Transaction fees are not new issuance. The current implementation pays all included fees to miners and does not burn them. Fees therefore redistribute existing ATHO from transaction senders to miners while subsidy expands scheduled supply. Demand, actual block timing, transaction volume, optional tips, lost keys, deliberate burns, and market price are not predictable from consensus constants and are separated from the deterministic model.

Keywords: Atho, monetary policy, supply model, tail emission, block subsidy, transaction fees, miner revenue, inflation, 500-year projection

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Executive Summary

- The genesis block at height 0 has zero subsidy and creates no premine.
- Heights 1 through 1,250,000 pay 8 ATHO per block.
- Heights 1,250,001 through 2,500,000 pay 4 ATHO per block.
- Heights 2,500,001 through 3,750,000 pay 2 ATHO per block.
- Heights 3,750,001 through 5,000,000 pay 1 ATHO per block.
- Height 5,000,001 and every later height pay 0.50 ATHO per block.
- Bootstrap issuance is exactly 18,750,000 ATHO.
- Atho has permanent tail emission and therefore no finite maximum supply.
- Expected tail issuance is 432 ATHO per day and 157,680 ATHO per year.
- Expected year-500 scheduled supply is 95,090,000 ATHO.
- Average gross issuance over the complete 500-year interval is 190,180 ATHO per year, but issuance is heavily front-loaded in the bootstrap eras.
- Target-cadence time-weighted average scheduled supply over the interval is approximately 55,526,315.0051 ATHO.
- The arithmetic mean of the 500 end-of-year supply checkpoints is 55,621,268.32 ATHO.
- The year-500 annualized issuance rate is approximately 0.165821853 percent.
- Every non-coinbase transaction pays at least $\max(600 \text{ atoms}, 1 \text{ atom per serialized byte})$.
- Any fee above the required floor is an optional miner tip.
- Included fees are miner revenue and are not burned.
- TX-PoW is an independent anti-spam requirement and does not replace fees.

1. Scope, Definitions, and Assumptions

1.1. What This Model Calculates

The deterministic portion of this model calculates scheduled block subsidy at each height and cumulative scheduled subsidy through selected heights. It also converts the 100-second target cadence into nominal days and years so that the height-based policy can be shown on a human time axis.

The complete annual appendix lists every year from 1 through 500. Each row shows the expected end-of-year height, the reward applicable to the following block, cumulative scheduled subsidy, next-year annualized subsidy at that reward, and the ratio of next-year issuance to end-of-year scheduled supply.

1.2. What This Model Does Not Predict

This model does not predict ATHO price, hashrate, transaction demand, fee tips, actual wall-clock block production, miner costs, lost private keys, deliberate burns, exchange liquidity, adoption, purchasing power, or legal treatment. Those outcomes depend on behavior and external conditions rather than the subsidy function.

1.3. Target-Cadence Convention

One nominal year is treated as 365 days:

seconds per block = 100
 blocks per day = $86,400 / 100 = 864$
 blocks per nominal year = $864 \times 365 = 315,360$
 expected end-of-year height = year $\times$ 315,360

Actual chain time can run ahead of or behind this projection over finite periods. Consensus rewards are selected by block height, not by calendar date.

1.4. Monetary Units

One ATHO contains 100,000,000 atoms and is displayed with eight decimal places. Consensus accounting uses integer atoms. Floating-point arithmetic does not determine valid rewards, outputs, or fees.

Table 1

Monetary Units and Minimum Values

Unit	Exact Value	Purpose
ATHO	100,000,000 atoms	Human-facing monetary unit
atom	0.00000001 ATHO	Smallest consensus accounting unit
dust threshold	100 atoms	Minimum standard output value
flat transaction fee floor	600 atoms	Minimum fee before the size floor is considered

2. Frozen Monetary Policy Constants

The values in this section are current repository rules. Reward validation uses the same values across mainnet, testnet, regnet, and prunetest; the networks remain isolated by their other consensus identities.

Table 2*Current Monetary Policy Constants*

Policy Element	Current Value	Consensus Meaning
Genesis height	0	Identity anchor with zero subsidy
First reward height	1	First ordinarily mineable subsidy block
Target block time	100 seconds	Expected cadence, not a timestamp reward trigger
Bootstrap era length	1,250,000 blocks	Fixed height span for each bootstrap reward
Bootstrap era count	4	Rewards of 8, 4, 2, and 1 ATHO
Bootstrap reward blocks	5,000,000	Total blocks before tail activation
Tail start height	5,000,001	First permanent 0.50 ATHO reward
Bootstrap supply	18,750,000 ATHO	Total subsidy through height 5,000,000
Tail reward	0.50 ATHO	Permanent subsidy for every later valid block
Premine	None	Genesis creates no spendable allocation
Maximum supply	None	Tail issuance does not stop
Coinbase maturity	100 blocks	Delay before a newly mined output may be spent

2.1. Reward Function

For block height h , the scheduled subsidy $R(h)$ is:

$$\begin{aligned}
 R(0) &= 0 \\
 R(h) &= 8.00 \text{ ATHO} && \text{for } 1 \leq h \leq 1,250,000 \\
 R(h) &= 4.00 \text{ ATHO} && \text{for } 1,250,001 \leq h \leq 2,500,000 \\
 R(h) &= 2.00 \text{ ATHO} && \text{for } 2,500,001 \leq h \leq 3,750,000 \\
 R(h) &= 1.00 \text{ ATHO} && \text{for } 3,750,001 \leq h \leq 5,000,000 \\
 R(h) &= 0.50 \text{ ATHO} && \text{for } h \geq 5,000,001
 \end{aligned}$$

The permanent tail is a floor, not a fifth temporary era. No later automatic halving or zero-subsidy cutoff exists in the current rules.

3. Bootstrap Eras and Tail Activation

At target cadence, one 1,250,000-block era lasts approximately 1,446.759 days, or 3.963724 nominal years. All dates below are target-cadence estimates; height remains authoritative.

Table 3*Reward Eras and Issuance*

Era	Heights and Reward	Issuance and Cumulative Supply
Genesis	Height 0 at 0 ATHO	0 issued; no premine
Era 1	1 to 1,250,000 at 8 ATHO	10,000,000 issued; 10,000,000 cumulative
Era 2	1,250,001 to 2,500,000 at 4 ATHO	5,000,000 issued; 15,000,000 cumulative
Era 3	2,500,001 to 3,750,000 at 2 ATHO	2,500,000 issued; 17,500,000 cumulative
Era 4	3,750,001 to 5,000,000 at 1 ATHO	1,250,000 issued; 18,750,000 cumulative
Tail	5,000,001 onward at 0.50 ATHO	157,680 expected per year without an end height

Table 4*Target-Cadence Era Milestones*

Milestone	Expected Nominal Age	Scheduled Supply
End Era 1	3.963724 years	10,000,000 ATHO
End Era 2	7.927448 years	15,000,000 ATHO
End Era 3	11.891172 years	17,500,000 ATHO
End Era 4	15.854896 years	18,750,000 ATHO
First tail block	approximately 15.8549 years	18,750,000.50 ATHO after inclusion

4. Cumulative Supply Mathematics

4.1. 500-Year Supply Curve

Target-cadence cumulative scheduled subsidy

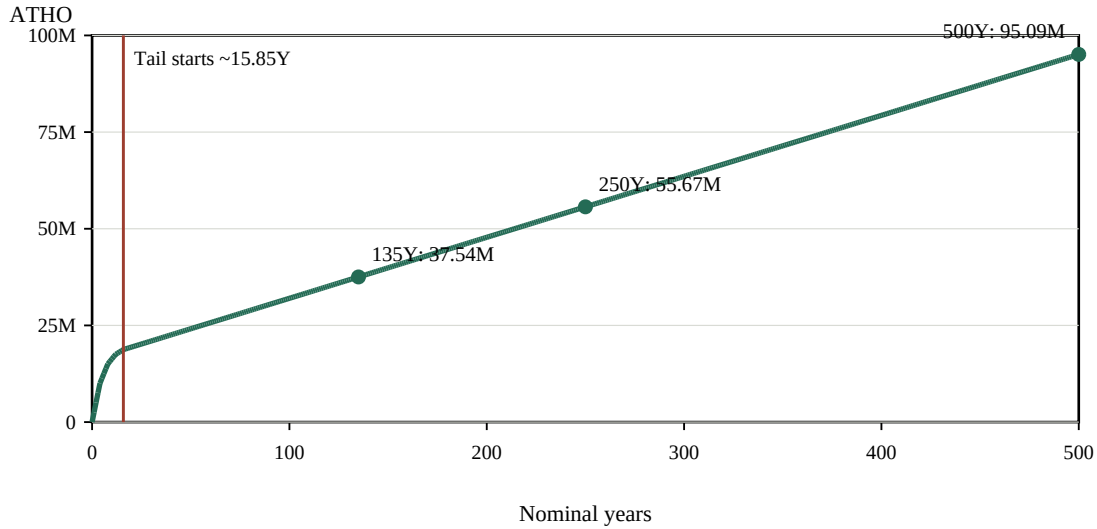


Figure 1. Cumulative Scheduled Supply

Let N be the number of reward-bearing blocks through a height. Cumulative scheduled subsidy $S(N)$ is the checked sum of the five reward regions:

$$\begin{aligned}
 S(N) = & 8.0 \times \min(N, 1,250,000) \\
 & + 4.0 \times \min(\max(N - 1,250,000, 0), 1,250,000) \\
 & + 2.0 \times \min(\max(N - 2,500,000, 0), 1,250,000) \\
 & + 1.0 \times \min(\max(N - 3,750,000, 0), 1,250,000) \\
 & + 0.5 \times \max(N - 5,000,000, 0)
 \end{aligned}$$

For integer nominal year y , this model uses $N = 315,360 \times y$. After tail activation the equation simplifies to:

$$S(N) = 18,750,000 + 0.50 \times (N - 5,000,000)$$

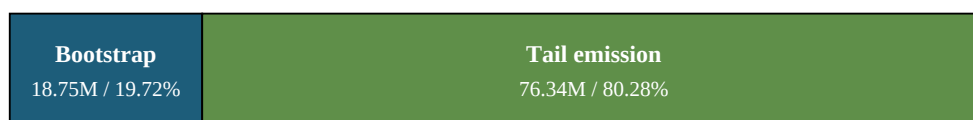
At year 500:

$$\begin{aligned}
 \text{expected reward-bearing blocks} &= 500 \times 315,360 = 157,680,000 \\
 \text{tail blocks} &= 157,680,000 - 5,000,000 = 152,680,000 \\
 \text{tail issuance} &= 152,680,000 \times 0.50 = 76,340,000 \text{ ATHO} \\
 \text{total scheduled supply} &= 18,750,000 + 76,340,000 \\
 &= 95,090,000 \text{ ATHO}
 \end{aligned}$$

4.2. Selected 500-Year Checkpoints

Table 5*Selected 500-Year Supply Checkpoints*

Year	End Height	Scheduled Supply
1	315,360	2,522,880 ATHO
5	1,576,800	11,307,200 ATHO
10	3,153,600	16,307,200 ATHO
15	4,730,400	18,480,400 ATHO
20	6,307,200	19,403,600 ATHO
25	7,884,000	20,192,000 ATHO
50	15,768,000	24,134,000 ATHO
75	23,652,000	28,076,000 ATHO
100	31,536,000	32,018,000 ATHO
135	42,573,600	37,536,800 ATHO
150	47,304,000	39,902,000 ATHO
200	63,072,000	47,786,000 ATHO
250	78,840,000	55,670,000 ATHO
300	94,608,000	63,554,000 ATHO
400	126,144,000	79,322,000 ATHO
500	157,680,000	95,090,000 ATHO

4.3. Composition at Year 500*Year-500 scheduled supply composition***Total scheduled subsidy: 95,090,000 ATHO***Figure 2. Bootstrap and Tail Composition at Year 500*

At year 500, bootstrap issuance is approximately 19.718162 percent of scheduled supply and tail issuance is approximately 80.281838 percent. Tail issuance becomes equal to the 18,750,000 ATHO bootstrap amount about 118.911720 years after tail activation, at a total chain age of approximately 134.766616 years. At that point, scheduled supply is approximately 37,500,000 ATHO and each source represents half.

4.4. Average Supply and Average Issuance

"Average supply over 500 years" can mean different things:

- Final scheduled supply is 95,090,000 ATHO.
- Gross issuance divided by 500 years is 190,180 ATHO per year.
- The target-cadence time-weighted average of the cumulative supply curve is approximately 55,526,315.0051 ATHO.
- The arithmetic mean of the 500 end-of-year supply values is 55,621,268.32 ATHO.

These are mathematical schedule averages. They are not estimates of circulating, liquid, or economically available supply.

5. Annual Issuance and Inflation

5.1. Issuance and Inflation Decline

Annualized issuance steps down; tail percentage keeps declining

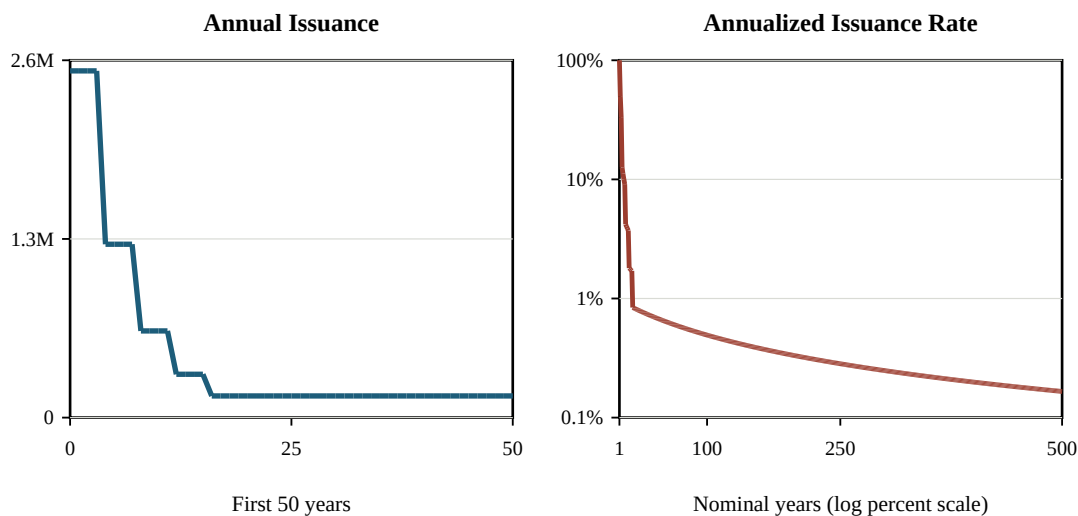


Figure 3. Annualized Issuance and Percentage Decline

Annualized issuance at a height is:

annualized issuance = reward for the next block x 315,360
 annualized issuance rate =
 annualized issuance / current scheduled supply x 100

Table 6*Daily and Annual Issuance by Reward*

Active Reward	Expected Daily Issuance	Expected Annual Issuance
8.00 ATHO	6,912 ATHO	2,522,880 ATHO
4.00 ATHO	3,456 ATHO	1,261,440 ATHO
2.00 ATHO	1,728 ATHO	630,720 ATHO
1.00 ATHO	864 ATHO	315,360 ATHO
0.50 ATHO	432 ATHO	157,680 ATHO

Table 7*Selected Annualized Issuance Rates*

Year	Scheduled Supply	Next-Year Issuance	Annualized Rate
1	2,522,880	2,522,880	100.000000000%
4	10,045,760	1,261,440	12.556939445%
5	11,307,200	1,261,440	11.156077544%
10	16,307,200	630,720	3.867739403%
15	18,480,400	315,360	1.706456570%
20	19,403,600	157,680	0.812632707%
50	24,134,000	157,680	0.653352117%
100	32,018,000	157,680	0.492472984%
250	55,670,000	157,680	0.283240525%
500	95,090,000	157,680	0.165821853%

5.2. Why Tail Inflation Declines Without Another Halving

The tail reward remains fixed in absolute terms at 0.50 ATHO per block. Scheduled supply continues to grow, so the same 157,680 ATHO annual increment becomes a smaller percentage of the existing base. The percentage approaches zero over a sufficiently long horizon but never becomes exactly zero while tail emission continues.

5.3. No Finite Maximum Supply

Atho does not converge to a fixed hard cap. After tail activation:

additional supply after t tail years = $157,680 \times t$ ATHO

This linear rule is intentional. It preserves a predictable base subsidy for miners rather than forcing a fee-only security budget at a fixed future height.

6. Supply Accounting Categories

6.1. Scheduled Issuance

Scheduled issuance is the subsidy permitted by consensus at each height. It does not include fees because fees already existed as transaction inputs before they were transferred to a miner.

6.2. Issued Supply

Issued supply is subsidy actually created in accepted blocks. At a perfectly on-schedule height it equals the model value. Calendar dates may differ from the nominal projection if block production differs from target cadence.

6.3. Circulating and Spendable Supply

Circulating supply is an interpretation rather than one consensus counter. New coinbase outputs are issued but remain immature for 100 blocks. Wallet and exchange policies can require additional confirmations without changing issuance.

6.4. Lost Coins

Coins controlled by lost private keys normally remain indistinguishable from dormant savings. Their percentage cannot be measured exactly from chainstate. No responsible supply report should subtract an assumed lost-coin percentage from scheduled supply and label the result as a consensus fact.

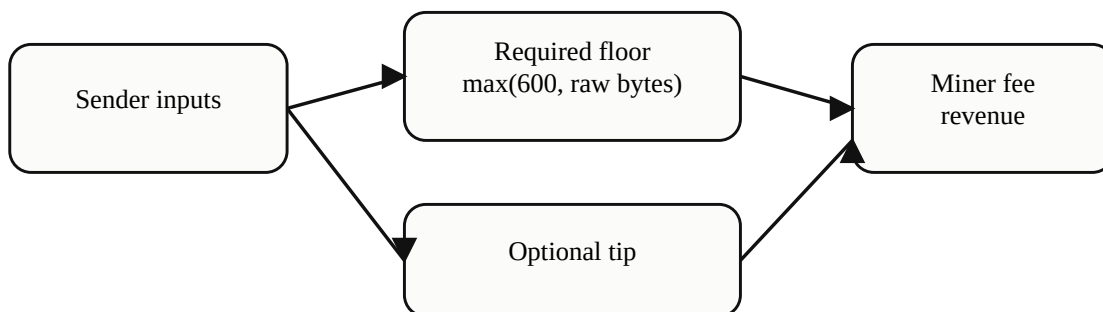
6.5. Burns and Unspendable Outputs

The protocol can identify deliberately unspendable constructions where their form is objectively known, but Atho does not burn transaction fees under the current policy. A deliberate burn reduces economically spendable supply but does not rewrite historical subsidy issuance.

7. Transaction Fee Policy

7.1. Fee Flow and Miner Payment

Transaction fees redistribute existing ATHO



No protocol fee burn; subsidy is the only new issuance in this flow

Figure 4. Transaction Fee Flow Under Current Policy

For every non-coinbase transaction:

transaction fee = total input value - total output value
 required fee = max(600 atoms, serialized transaction bytes × 1 atom)

optional tip = transaction fee - required fee
 valid only when transaction fee $\geq$ required fee
 miner may claim active subsidy + all included transaction fees

The 600-atom amount is a minimum, not a maximum. For transactions larger than 600 serialized bytes, the one-atom-per-byte rule is the binding floor.

7.2. Canonical Transaction Shape Examples

These sizes come from the repository's canonical Falcon transaction benchmark. They are useful reference shapes, not guarantees about every wallet-created transaction.

Table 8

Canonical Transaction Fee Examples

Transaction Shape	Raw Bytes / Vbytes	Required Floor Fee
Single payment	1,730 / 518	1,730 atoms = 0.00001730 ATHO
Payment with change	1,774 / 562	1,774 atoms = 0.00001774 ATHO
Small consolidation	2,008 / 747	2,008 atoms = 0.00002008 ATHO
Maximum-output batch	4,502 / 3,290	4,502 atoms = 0.00004502 ATHO
Maximum input/output shape	84,296 / 66,205	84,296 atoms = 0.00084296 ATHO
Large 128-key consolidation	211,705 / 58,379	211,705 atoms = 0.00211705 ATHO

7.3. Optional Tips

Wallets may add a tip above the required floor. A tip can improve miner priority during congestion, but it is not required for consensus validity when the transaction already satisfies the floor. Miner selection policy and mempool conditions determine whether a higher fee changes inclusion time.

7.4. TX-PoW Is Separate

Every ordinary transaction also carries deterministic 10-to-16-bit TX-PoW based on transaction shape. TX-PoW is a pass/fail anti-spam gate:

- it does not create or destroy ATHO
- it does not pay miners
- it does not replace the fee floor
- extra work does not buy priority
- it does not add proof of work to the blockchain tip

8. Fee Value at Illustrative ATHO Prices

This table is a conversion tool, not a price prediction. USD fee value equals fee in ATHO $\times$ assumed ATHO price.

Table 9*Illustrative Fiat Fee Conversion*

Fee Example	At \$0.01 / ATHO	At \$1 / ATHO	At \$100 / ATHO	At \$1,000 / ATHO
Flat 600-atom floor	\$0.00000006	\$0.00000600	\$0.00060000	\$0.00600000
1,730-atom single payment	\$0.000000173	\$0.00001730	\$0.00173000	\$0.01730000
1,774-atom payment with change	\$0.0000001774	\$0.00001774	\$0.00177400	\$0.01774000
4,502-atom output batch	\$0.0000004502	\$0.00004502	\$0.00450200	\$0.04502000
84,296-atom maximum shape	\$0.0000084296	\$0.00084296	\$0.08429600	\$0.84296000

Because the fee floor is denominated in ATHO rather than fiat, its fiat value moves with the market value of ATHO. Wallet fee recommendations can react to mempool demand, but changing the consensus fee formula would require a coordinated consensus upgrade.

9. Adaptive Block Capacity and Fee Throughput

9.1. Block Capacity and Full-Block Fee Scenarios

Full-block fee revenue as a percentage of 0.50 ATHO tail subsidy

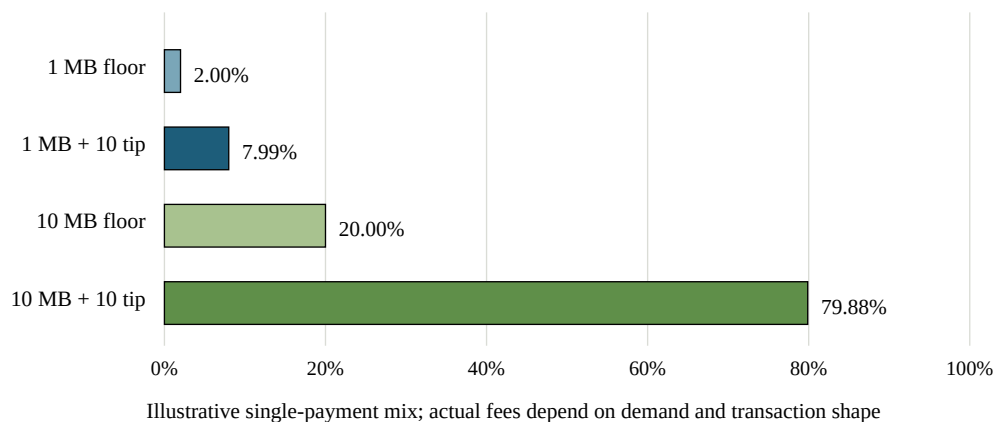


Figure 5. Full-Block Fee Revenue Relative to Tail Subsidy

Atho launches with an active 1,000,000-byte block limit. The limit is evaluated in 864-block epochs:

- usage at or above 85 percent expands the next epoch limit
- usage at or below 35 percent shrinks the next epoch limit
- usage between those thresholds holds the limit

- sustained expansion reaches the 10,000,000-byte maximum over 9,125 daily epochs, or 25 nominal years
- sustained shrinkage can traverse the range over 1,825 daily epochs, or 5 nominal years

The network also enforces hard ceilings of 10,000,000 vbytes, 40,000,000 weight units, and 10,000,000 raw serialized bytes.

9.2. Single-Payment Full-Block Model

The following model fills the raw-byte budget with the 1,730-raw-byte, 518-vbyte single-payment benchmark shape. It ignores coinbase and block overhead, so it is a payload-only ceiling rather than an exact template guarantee.

Table 10

Single-Payment Payload Capacity

Active Raw Limit	Approx. Payments	Used Raw Bytes / Vbytes
1,000,000 bytes	578	999,940 / 299,404
10,000,000 bytes	5,780	9,999,400 / 2,994,040

Table 11

Full-Block Fee Scenarios

Limit and Tip Rate	Fee per Full Block	Annual Fees at 315,360 Blocks
1 MB, floor only	0.00999940 ATHO	3,153.410784 ATHO
1 MB, plus 1 atom/vbyte	0.01299344 ATHO	4,097.611238 ATHO
1 MB, plus 5 atoms/vbyte	0.02496960 ATHO	7,874.413056 ATHO
1 MB, plus 10 atoms/vbyte	0.03993980 ATHO	12,595.415328 ATHO
10 MB, floor only	0.09999400 ATHO	31,534.107840 ATHO
10 MB, plus 1 atom/vbyte	0.12993440 ATHO	40,976.112384 ATHO
10 MB, plus 5 atoms/vbyte	0.24969600 ATHO	78,744.130560 ATHO
10 MB, plus 10 atoms/vbyte	0.39939800 ATHO	125,954.153280 ATHO

Real fee totals depend on transaction mix, witness sharing, coinbase and block overhead, optional tips, miner selection, and whether demand fills every block. An empty block carries no transaction fees.

9.3. Fee Revenue Relative to Subsidy

Table 12*Floor Fees Relative to Subsidy*

Reward Era	1 MB Floor-Fee Block	10 MB Floor-Fee Block
8 ATHO subsidy	about 0.125% of subsidy	about 1.250% of subsidy
4 ATHO subsidy	about 0.250% of subsidy	about 2.500% of subsidy
2 ATHO subsidy	about 0.500% of subsidy	about 5.000% of subsidy
1 ATHO subsidy	about 1.000% of subsidy	about 9.999% of subsidy
0.50 ATHO tail	about 2.000% of subsidy	about 19.999% of subsidy

These percentages describe full-block floor fees as miner revenue. They do not describe net supply reduction because current fees are not burned.

10. Miner Revenue Composition

10.1. Subsidy and Fee Components

For a valid block:

maximum allowed coinbase value =
active block subsidy + sum of included transaction fees

Miners cannot claim more than that checked amount. Subsidy is newly issued ATHO. Fees are existing ATHO transferred from users. Optional tips are included in the same fee total.

10.2. Tail-Era Security Budget

The permanent tail provides a deterministic base revenue of 0.50 ATHO per valid block before fees. At target cadence that is 432 ATHO per day and 157,680 ATHO per year across all miners.

Fees can supplement that base:

tail-era miner revenue per block = 0.50 ATHO + included fees

The rule does not guarantee profitable mining. Fiat revenue depends on ATHO price, hash competition, hardware, electricity, pool terms, uptime, and actual fee demand.

11. Fee Burning Counterfactual

11.1. Current Policy: No Fee Burn

Under current rules:

sender balance decreases by payment outputs + fee
recipient balances increase by payment outputs
miner balance increases by subsidy + fee
scheduled supply increases only by subsidy

The fee moves existing units and therefore neither offsets nor increases scheduled issuance.

11.2. What Full Fee Burning Would Change

Burning fees would require a consensus and accounting redesign. As a strictly hypothetical comparison, if all full-block fees in the single-payment model were burned:

- 1 MB floor-only blocks would burn about 3,153.41 ATHO per year, offsetting about 2.00 percent of 157,680 ATHO annual tail issuance
- 10 MB floor-only blocks would burn about 31,534.11 ATHO per year, offsetting about 20.00 percent of annual tail issuance
- 10 MB blocks with a 10-atom/vbyte optional tip would burn about 125,954.15 ATHO per year, offsetting about 79.88 percent of annual tail issuance

At the one-atom-per-row-byte floor, fully offsetting the 0.50 ATHO tail would require 50,000,000 fee-bearing raw bytes per block. That exceeds the current 10,000,000-row-byte hard ceiling. Optional tips could raise fees without more bytes, but tips and sustained full blocks cannot be assumed.

This counterfactual is not active Atho policy and is not included in any supply total in this document.

12. Monetary Validation and Safety

12.1. Block Accounting Flow

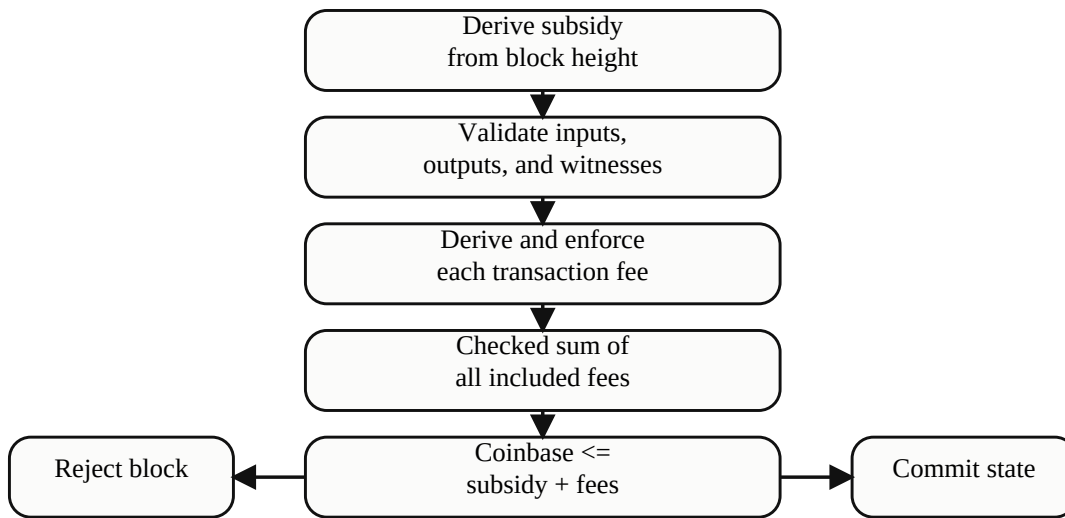


Figure 6. Consensus Monetary Validation Flow

Full nodes independently:

- 1 derive the active subsidy from the candidate block height
- 2 validate each non-coinbase transaction and derive its fee from inputs minus outputs
- 3 enforce the required fee floor for every included transaction
- 4 sum all included fees with checked integer arithmetic
- 5 reject duplicate spends, missing inputs, invalid witnesses, and value creation
- 6 reject a coinbase that claims more than subsidy plus validated fees
- 7 commit UTXO and block state only after all checks pass

12.2. Coinbase Maturity

Coinbase outputs require 100 blocks of maturity. At target cadence that is approximately 10,000 seconds, or 2 hours 46 minutes 40 seconds. Maturity delays spendability but does not delay the accounting classification of subsidy as issued.

12.3. Integer Arithmetic

Consensus values use atoms and checked integer arithmetic. Human decimal formatting is presentation only. The use of eight decimal places does not imply floating-point consensus calculations.

12.4. Changing the Policy

The subsidy schedule, fee floor, monetary unit, and coinbase accounting are consensus-critical. Changing them after network launch requires an explicitly specified activation and coordinated consensus upgrade. A documentation edit, wallet preference, miner configuration, or API field cannot change these rules.

13. Economic Interpretation

13.1. Strengths of the Current Model

- The first 18,750,000 ATHO is delivered through simple, auditable eras.
- The permanent 0.50 ATHO tail avoids a sudden transition to fee-only mining.
- Tail issuance is fixed in absolute terms, so its percentage rate declines as scheduled supply grows.
- Fees remain available as an additional miner-security component.
- No premine or hidden genesis allocation exists.
- The exact reward at any height can be calculated without market data.

13.2. Tradeoffs

- There is no finite maximum supply.
- Long-run nominal supply grows linearly.
- The fiat value of the fee floor is not stable because the floor is denominated in ATHO.
- A tail reward does not guarantee enough hashrate or decentralization.
- A low fee floor requires mempool, peer-rate, TX-PoW, and resource controls to remain effective against spam.
- Lost coins can make economically available supply lower than scheduled supply, but the difference is not exactly measurable.

13.3. Reading the 500-Year Number Correctly

95,090,000 ATHO is the target-cadence scheduled subsidy at year 500. It is not a hard cap, a circulating-supply promise, a price target, or a statement that the network will survive for 500 years. The value is a transparent consequence of the current height-based reward function.

14. Reproduction and Code References

The model is grounded in these repository locations:

Table 13*Monetary Model Code References*

Subject	Authoritative Code
Monetary units, rewards, cadence, fees, dust, and block limits	<code>crates/atho-core/src/constants.rs</code>
Reward and cumulative issuance functions	<code>crates/atho-core/src/consensus/subsidy.rs</code>
Required transaction fee calculation	<code>crates/atho-core/src/consensus/tx_policy.rs</code>
Adaptive block limit	<code>crates/atho-core/src/consensus/block_size.rs</code>
Coinbase and block monetary validation	<code>crates/atho-storage/src/chainstate.rs</code>
Canonical transaction size examples	<code>docs/TX_POW_BENCHMARK.md</code>

Rebuild this PDF with:

```
python3 scripts/render_monetary_supply_model.py
```

The renderer validates its core constants against the Rust source before calculating charts and the complete annual appendix.

Appendix A. Annual Schedule, Years 1-500

The PDF version appends a generated 500-row schedule. Each row is calculated from the current repository constants rather than manually transcribed. The table uses the following column definitions:

- **Year:** nominal 365-day target-cadence year
- **End Height:** expected reward-bearing block height at year end
- **Next Reward:** subsidy for the next expected block
- **Scheduled Supply:** cumulative subsidy through end height
- **Next-Year Issuance:** next reward multiplied by 315,360
- **Annualized Rate:** next-year issuance divided by end-of-year scheduled supply

Table 14

Complete Target-Cadence Annual Supply Schedule

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
1	315,360	8.00	2,522,880	2,522,880	100.000000000%
2	630,720	8.00	5,045,760	2,522,880	50.000000000%
3	946,080	8.00	7,568,640	2,522,880	33.333333333%
4	1,261,440	4.00	10,045,760	1,261,440	12.556939445%
5	1,576,800	4.00	11,307,200	1,261,440	11.156077544%
6	1,892,160	4.00	12,568,640	1,261,440	10.036408076%
7	2,207,520	4.00	13,830,080	1,261,440	9.120988454%
8	2,522,880	2.00	15,045,760	630,720	4.192011570%
9	2,838,240	2.00	15,676,480	630,720	4.023352181%
10	3,153,600	2.00	16,307,200	630,720	3.867739403%
11	3,468,960	2.00	16,937,920	630,720	3.723715781%
12	3,784,320	1.00	17,534,320	315,360	1.798529969%
13	4,099,680	1.00	17,849,680	315,360	1.766754362%
14	4,415,040	1.00	18,165,040	315,360	1.736082057%
15	4,730,400	1.00	18,480,400	315,360	1.706456570%
16	5,045,760	0.50	18,772,880	157,680	0.839935055%
17	5,361,120	0.50	18,930,560	157,680	0.832938909%
18	5,676,480	0.50	19,088,240	157,680	0.826058348%
19	5,991,840	0.50	19,245,920	157,680	0.819290530%
20	6,307,200	0.50	19,403,600	157,680	0.812632707%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
21	6,622,560	0.50	19,561,280	157,680	0.806082220%
22	6,937,920	0.50	19,718,960	157,680	0.799636492%
23	7,253,280	0.50	19,876,640	157,680	0.793293031%
24	7,568,640	0.50	20,034,320	157,680	0.787049423%
25	7,884,000	0.50	20,192,000	157,680	0.780903328%
26	8,199,360	0.50	20,349,680	157,680	0.774852479%
27	8,514,720	0.50	20,507,360	157,680	0.768894680%
28	8,830,080	0.50	20,665,040	157,680	0.763027800%
29	9,145,440	0.50	20,822,720	157,680	0.757249773%
30	9,460,800	0.50	20,980,400	157,680	0.751558598%
31	9,776,160	0.50	21,138,080	157,680	0.745952329%
32	10,091,520	0.50	21,295,760	157,680	0.740429081%
33	10,406,880	0.50	21,453,440	157,680	0.734987023%
34	10,722,240	0.50	21,611,120	157,680	0.729624379%
35	11,037,600	0.50	21,768,800	157,680	0.724339422%
36	11,352,960	0.50	21,926,480	157,680	0.719130476%
37	11,668,320	0.50	22,084,160	157,680	0.713995914%
38	11,983,680	0.50	22,241,840	157,680	0.708934153%
39	12,299,040	0.50	22,399,520	157,680	0.703943656%
40	12,614,400	0.50	22,557,200	157,680	0.699022928%
41	12,929,760	0.50	22,714,880	157,680	0.694170517%
42	13,245,120	0.50	22,872,560	157,680	0.689385010%
43	13,560,480	0.50	23,030,240	157,680	0.684665032%
44	13,875,840	0.50	23,187,920	157,680	0.680009246%
45	14,191,200	0.50	23,345,600	157,680	0.675416353%
46	14,506,560	0.50	23,503,280	157,680	0.670885085%
47	14,821,920	0.50	23,660,960	157,680	0.666414211%
48	15,137,280	0.50	23,818,640	157,680	0.662002532%
49	15,452,640	0.50	23,976,320	157,680	0.657648880%
50	15,768,000	0.50	24,134,000	157,680	0.653352117%
51	16,083,360	0.50	24,291,680	157,680	0.649111136%
52	16,398,720	0.50	24,449,360	157,680	0.644924857%
53	16,714,080	0.50	24,607,040	157,680	0.640792229%
54	17,029,440	0.50	24,764,720	157,680	0.636712226%
55	17,344,800	0.50	24,922,400	157,680	0.632683851%
56	17,660,160	0.50	25,080,080	157,680	0.628706129%
57	17,975,520	0.50	25,237,760	157,680	0.624778110%
58	18,290,880	0.50	25,395,440	157,680	0.620898870%
59	18,606,240	0.50	25,553,120	157,680	0.617067505%
60	18,921,600	0.50	25,710,800	157,680	0.613283134%
61	19,236,960	0.50	25,868,480	157,680	0.609544898%
62	19,552,320	0.50	26,026,160	157,680	0.605851958%
63	19,867,680	0.50	26,183,840	157,680	0.602203497%
64	20,183,040	0.50	26,341,520	157,680	0.598598714%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
65	20,498,400	0.50	26,499,200	157,680	0.595036831%
66	20,813,760	0.50	26,656,880	157,680	0.591517087%
67	21,129,120	0.50	26,814,560	157,680	0.588038737%
68	21,444,480	0.50	26,972,240	157,680	0.584601056%
69	21,759,840	0.50	27,129,920	157,680	0.581203336%
70	22,075,200	0.50	27,287,600	157,680	0.577844882%
71	22,390,560	0.50	27,445,280	157,680	0.574525019%
72	22,705,920	0.50	27,602,960	157,680	0.571243084%
73	23,021,280	0.50	27,760,640	157,680	0.567998432%
74	23,336,640	0.50	27,918,320	157,680	0.564790432%
75	23,652,000	0.50	28,076,000	157,680	0.561618464%
76	23,967,360	0.50	28,233,680	157,680	0.558481927%
77	24,282,720	0.50	28,391,360	157,680	0.555380228%
78	24,598,080	0.50	28,549,040	157,680	0.552312792%
79	24,913,440	0.50	28,706,720	157,680	0.549279054%
80	25,228,800	0.50	28,864,400	157,680	0.546278461%
81	25,544,160	0.50	29,022,080	157,680	0.543310473%
82	25,859,520	0.50	29,179,760	157,680	0.540374561%
83	26,174,880	0.50	29,337,440	157,680	0.537470209%
84	26,490,240	0.50	29,495,120	157,680	0.534596910%
85	26,805,600	0.50	29,652,800	157,680	0.531754168%
86	27,120,960	0.50	29,810,480	157,680	0.528941500%
87	27,436,320	0.50	29,968,160	157,680	0.526158429%
88	27,751,680	0.50	30,125,840	157,680	0.523404493%
89	28,067,040	0.50	30,283,520	157,680	0.520679234%
90	28,382,400	0.50	30,441,200	157,680	0.517982208%
91	28,697,760	0.50	30,598,880	157,680	0.515312979%
92	29,013,120	0.50	30,756,560	157,680	0.512671118%
93	29,328,480	0.50	30,914,240	157,680	0.510056207%
94	29,643,840	0.50	31,071,920	157,680	0.507467836%
95	29,959,200	0.50	31,229,600	157,680	0.504905602%
96	30,274,560	0.50	31,387,280	157,680	0.502369113%
97	30,589,920	0.50	31,544,960	157,680	0.499857980%
98	30,905,280	0.50	31,702,640	157,680	0.497371828%
99	31,220,640	0.50	31,860,320	157,680	0.494910283%
100	31,536,000	0.50	32,018,000	157,680	0.492472984%
101	31,851,360	0.50	32,175,680	157,680	0.490059573%
102	32,166,720	0.50	32,333,360	157,680	0.487669701%
103	32,482,080	0.50	32,491,040	157,680	0.485303025%
104	32,797,440	0.50	32,648,720	157,680	0.482959209%
105	33,112,800	0.50	32,806,400	157,680	0.480637924%
106	33,428,160	0.50	32,964,080	157,680	0.478338846%
107	33,743,520	0.50	33,121,760	157,680	0.476061659%
108	34,058,880	0.50	33,279,440	157,680	0.473806050%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
109	34,374,240	0.50	33,437,120	157,680	0.471571714%
110	34,689,600	0.50	33,594,800	157,680	0.469358353%
111	35,004,960	0.50	33,752,480	157,680	0.467165672%
112	35,320,320	0.50	33,910,160	157,680	0.464993383%
113	35,635,680	0.50	34,067,840	157,680	0.462841202%
114	35,951,040	0.50	34,225,520	157,680	0.460708851%
115	36,266,400	0.50	34,383,200	157,680	0.458596059%
116	36,581,760	0.50	34,540,880	157,680	0.456502556%
117	36,897,120	0.50	34,698,560	157,680	0.454428080%
118	37,212,480	0.50	34,856,240	157,680	0.452372373%
119	37,527,840	0.50	35,013,920	157,680	0.450335181%
120	37,843,200	0.50	35,171,600	157,680	0.448316255%
121	38,158,560	0.50	35,329,280	157,680	0.446315351%
122	38,473,920	0.50	35,486,960	157,680	0.444332228%
123	38,789,280	0.50	35,644,640	157,680	0.442366650%
124	39,104,640	0.50	35,802,320	157,680	0.440418386%
125	39,420,000	0.50	35,960,000	157,680	0.438487208%
126	39,735,360	0.50	36,117,680	157,680	0.436572892%
127	40,050,720	0.50	36,275,360	157,680	0.434675218%
128	40,366,080	0.50	36,433,040	157,680	0.432793969%
129	40,681,440	0.50	36,590,720	157,680	0.430928935%
130	40,996,800	0.50	36,748,400	157,680	0.429079906%
131	41,312,160	0.50	36,906,080	157,680	0.427246676%
132	41,627,520	0.50	37,063,760	157,680	0.425429044%
133	41,942,880	0.50	37,221,440	157,680	0.423626813%
134	42,258,240	0.50	37,379,120	157,680	0.421839786%
135	42,573,600	0.50	37,536,800	157,680	0.420067773%
136	42,888,960	0.50	37,694,480	157,680	0.418310586%
137	43,204,320	0.50	37,852,160	157,680	0.416568037%
138	43,519,680	0.50	38,009,840	157,680	0.414839947%
139	43,835,040	0.50	38,167,520	157,680	0.413126134%
140	44,150,400	0.50	38,325,200	157,680	0.411426424%
141	44,465,760	0.50	38,482,880	157,680	0.409740643%
142	44,781,120	0.50	38,640,560	157,680	0.408068620%
143	45,096,480	0.50	38,798,240	157,680	0.406410188%
144	45,411,840	0.50	38,955,920	157,680	0.404765181%
145	45,727,200	0.50	39,113,600	157,680	0.403133437%
146	46,042,560	0.50	39,271,280	157,680	0.401514797%
147	46,357,920	0.50	39,428,960	157,680	0.399909102%
148	46,673,280	0.50	39,586,640	157,680	0.398316200%
149	46,988,640	0.50	39,744,320	157,680	0.396735936%
150	47,304,000	0.50	39,902,000	157,680	0.395168162%
151	47,619,360	0.50	40,059,680	157,680	0.393612730%
152	47,934,720	0.50	40,217,360	157,680	0.392069494%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
153	48,250,080	0.50	40,375,040	157,680	0.390538313%
154	48,565,440	0.50	40,532,720	157,680	0.389019044%
155	48,880,800	0.50	40,690,400	157,680	0.387511551%
156	49,196,160	0.50	40,848,080	157,680	0.386015695%
157	49,511,520	0.50	41,005,760	157,680	0.384531344%
158	49,826,880	0.50	41,163,440	157,680	0.383058364%
159	50,142,240	0.50	41,321,120	157,680	0.381596627%
160	50,457,600	0.50	41,478,800	157,680	0.380146002%
161	50,772,960	0.50	41,636,480	157,680	0.378706365%
162	51,088,320	0.50	41,794,160	157,680	0.377277591%
163	51,403,680	0.50	41,951,840	157,680	0.375859557%
164	51,719,040	0.50	42,109,520	157,680	0.374452143%
165	52,034,400	0.50	42,267,200	157,680	0.373055230%
166	52,349,760	0.50	42,424,880	157,680	0.371668700%
167	52,665,120	0.50	42,582,560	157,680	0.370292439%
168	52,980,480	0.50	42,740,240	157,680	0.368926333%
169	53,295,840	0.50	42,897,920	157,680	0.367570269%
170	53,611,200	0.50	43,055,600	157,680	0.366224138%
171	53,926,560	0.50	43,213,280	157,680	0.364887831%
172	54,241,920	0.50	43,370,960	157,680	0.363561240%
173	54,557,280	0.50	43,528,640	157,680	0.362244260%
174	54,872,640	0.50	43,686,320	157,680	0.360936788%
175	55,188,000	0.50	43,844,000	157,680	0.359638719%
176	55,503,360	0.50	44,001,680	157,680	0.358349954%
177	55,818,720	0.50	44,159,360	157,680	0.357070392%
178	56,134,080	0.50	44,317,040	157,680	0.355799936%
179	56,449,440	0.50	44,474,720	157,680	0.354538488%
180	56,764,800	0.50	44,632,400	157,680	0.353285954%
181	57,080,160	0.50	44,790,080	157,680	0.352042238%
182	57,395,520	0.50	44,947,760	157,680	0.350807248%
183	57,710,880	0.50	45,105,440	157,680	0.349580893%
184	58,026,240	0.50	45,263,120	157,680	0.348363082%
185	58,341,600	0.50	45,420,800	157,680	0.347153727%
186	58,656,960	0.50	45,578,480	157,680	0.345952739%
187	58,972,320	0.50	45,736,160	157,680	0.344760032%
188	59,287,680	0.50	45,893,840	157,680	0.343575521%
189	59,603,040	0.50	46,051,520	157,680	0.342399122%
190	59,918,400	0.50	46,209,200	157,680	0.341230751%
191	60,233,760	0.50	46,366,880	157,680	0.340070326%
192	60,549,120	0.50	46,524,560	157,680	0.338917767%
193	60,864,480	0.50	46,682,240	157,680	0.337772995%
194	61,179,840	0.50	46,839,920	157,680	0.336635929%
195	61,495,200	0.50	46,997,600	157,680	0.335506494%
196	61,810,560	0.50	47,155,280	157,680	0.334384612%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
197	62,125,920	0.50	47,312,960	157,680	0.333270208%
198	62,441,280	0.50	47,470,640	157,680	0.332163207%
199	62,756,640	0.50	47,628,320	157,680	0.331063535%
200	63,072,000	0.50	47,786,000	157,680	0.329971121%
201	63,387,360	0.50	47,943,680	157,680	0.328885893%
202	63,702,720	0.50	48,101,360	157,680	0.327807779%
203	64,018,080	0.50	48,259,040	157,680	0.326736711%
204	64,333,440	0.50	48,416,720	157,680	0.325672619%
205	64,648,800	0.50	48,574,400	157,680	0.324615435%
206	64,964,160	0.50	48,732,080	157,680	0.323565093%
207	65,279,520	0.50	48,889,760	157,680	0.322521526%
208	65,594,880	0.50	49,047,440	157,680	0.321484669%
209	65,910,240	0.50	49,205,120	157,680	0.320454457%
210	66,225,600	0.50	49,362,800	157,680	0.319430826%
211	66,540,960	0.50	49,520,480	157,680	0.318413715%
212	66,856,320	0.50	49,678,160	157,680	0.317403060%
213	67,171,680	0.50	49,835,840	157,680	0.316398801%
214	67,487,040	0.50	49,993,520	157,680	0.315400876%
215	67,802,400	0.50	50,151,200	157,680	0.314409226%
216	68,117,760	0.50	50,308,880	157,680	0.313423793%
217	68,433,120	0.50	50,466,560	157,680	0.312444518%
218	68,748,480	0.50	50,624,240	157,680	0.311471343%
219	69,063,840	0.50	50,781,920	157,680	0.310504211%
220	69,379,200	0.50	50,939,600	157,680	0.309543067%
221	69,694,560	0.50	51,097,280	157,680	0.308587854%
222	70,009,920	0.50	51,254,960	157,680	0.307638519%
223	70,325,280	0.50	51,412,640	157,680	0.306695007%
224	70,640,640	0.50	51,570,320	157,680	0.305757265%
225	70,956,000	0.50	51,728,000	157,680	0.304825240%
226	71,271,360	0.50	51,885,680	157,680	0.303898879%
227	71,586,720	0.50	52,043,360	157,680	0.302978132%
228	71,902,080	0.50	52,201,040	157,680	0.302062947%
229	72,217,440	0.50	52,358,720	157,680	0.301153275%
230	72,532,800	0.50	52,516,400	157,680	0.300249065%
231	72,848,160	0.50	52,674,080	157,680	0.299350269%
232	73,163,520	0.50	52,831,760	157,680	0.298456837%
233	73,478,880	0.50	52,989,440	157,680	0.297568723%
234	73,794,240	0.50	53,147,120	157,680	0.296685879%
235	74,109,600	0.50	53,304,800	157,680	0.295808257%
236	74,424,960	0.50	53,462,480	157,680	0.294935813%
237	74,740,320	0.50	53,620,160	157,680	0.294068500%
238	75,055,680	0.50	53,777,840	157,680	0.293206272%
239	75,371,040	0.50	53,935,520	157,680	0.292349086%
240	75,686,400	0.50	54,093,200	157,680	0.291496898%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
241	76,001,760	0.50	54,250,880	157,680	0.290649663%
242	76,317,120	0.50	54,408,560	157,680	0.289807339%
243	76,632,480	0.50	54,566,240	157,680	0.288969883%
244	76,947,840	0.50	54,723,920	157,680	0.288137253%
245	77,263,200	0.50	54,881,600	157,680	0.287309408%
246	77,578,560	0.50	55,039,280	157,680	0.286486306%
247	77,893,920	0.50	55,196,960	157,680	0.285667906%
248	78,209,280	0.50	55,354,640	157,680	0.284854169%
249	78,524,640	0.50	55,512,320	157,680	0.284045055%
250	78,840,000	0.50	55,670,000	157,680	0.283240525%
251	79,155,360	0.50	55,827,680	157,680	0.282440538%
252	79,470,720	0.50	55,985,360	157,680	0.281645059%
253	79,786,080	0.50	56,143,040	157,680	0.280854047%
254	80,101,440	0.50	56,300,720	157,680	0.280067466%
255	80,416,800	0.50	56,458,400	157,680	0.279285279%
256	80,732,160	0.50	56,616,080	157,680	0.278507449%
257	81,047,520	0.50	56,773,760	157,680	0.277733939%
258	81,362,880	0.50	56,931,440	157,680	0.276964714%
259	81,678,240	0.50	57,089,120	157,680	0.276199738%
260	81,993,600	0.50	57,246,800	157,680	0.275438977%
261	82,308,960	0.50	57,404,480	157,680	0.274682394%
262	82,624,320	0.50	57,562,160	157,680	0.273929957%
263	82,939,680	0.50	57,719,840	157,680	0.273181630%
264	83,255,040	0.50	57,877,520	157,680	0.272437382%
265	83,570,400	0.50	58,035,200	157,680	0.271697177%
266	83,885,760	0.50	58,192,880	157,680	0.270960984%
267	84,201,120	0.50	58,350,560	157,680	0.270228769%
268	84,516,480	0.50	58,508,240	157,680	0.269500501%
269	84,831,840	0.50	58,665,920	157,680	0.268776148%
270	85,147,200	0.50	58,823,600	157,680	0.268055678%
271	85,462,560	0.50	58,981,280	157,680	0.267339061%
272	85,777,920	0.50	59,138,960	157,680	0.266626265%
273	86,093,280	0.50	59,296,640	157,680	0.265917259%
274	86,408,640	0.50	59,454,320	157,680	0.265212015%
275	86,724,000	0.50	59,612,000	157,680	0.264510501%
276	87,039,360	0.50	59,769,680	157,680	0.263812689%
277	87,354,720	0.50	59,927,360	157,680	0.263118549%
278	87,670,080	0.50	60,085,040	157,680	0.262428052%
279	87,985,440	0.50	60,242,720	157,680	0.261741170%
280	88,300,800	0.50	60,400,400	157,680	0.261057874%
281	88,616,160	0.50	60,558,080	157,680	0.260378136%
282	88,931,520	0.50	60,715,760	157,680	0.259701929%
283	89,246,880	0.50	60,873,440	157,680	0.259029225%
284	89,562,240	0.50	61,031,120	157,680	0.258359997%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
285	89,877,600	0.50	61,188,800	157,680	0.257694219%
286	90,192,960	0.50	61,346,480	157,680	0.257031862%
287	90,508,320	0.50	61,504,160	157,680	0.256372902%
288	90,823,680	0.50	61,661,840	157,680	0.255717312%
289	91,139,040	0.50	61,819,520	157,680	0.255065067%
290	91,454,400	0.50	61,977,200	157,680	0.254416140%
291	91,769,760	0.50	62,134,880	157,680	0.253770507%
292	92,085,120	0.50	62,292,560	157,680	0.253128142%
293	92,400,480	0.50	62,450,240	157,680	0.252489022%
294	92,715,840	0.50	62,607,920	157,680	0.251853120%
295	93,031,200	0.50	62,765,600	157,680	0.251220414%
296	93,346,560	0.50	62,923,280	157,680	0.250590878%
297	93,661,920	0.50	63,080,960	157,680	0.249964490%
298	93,977,280	0.50	63,238,640	157,680	0.249341226%
299	94,292,640	0.50	63,396,320	157,680	0.248721061%
300	94,608,000	0.50	63,554,000	157,680	0.248103975%
301	94,923,360	0.50	63,711,680	157,680	0.247489942%
302	95,238,720	0.50	63,869,360	157,680	0.246878942%
303	95,554,080	0.50	64,027,040	157,680	0.246270951%
304	95,869,440	0.50	64,184,720	157,680	0.245665947%
305	96,184,800	0.50	64,342,400	157,680	0.245063908%
306	96,500,160	0.50	64,500,080	157,680	0.244464813%
307	96,815,520	0.50	64,657,760	157,680	0.243868640%
308	97,130,880	0.50	64,815,440	157,680	0.243275368%
309	97,446,240	0.50	64,973,120	157,680	0.242684975%
310	97,761,600	0.50	65,130,800	157,680	0.242097441%
311	98,076,960	0.50	65,288,480	157,680	0.241512745%
312	98,392,320	0.50	65,446,160	157,680	0.240930866%
313	98,707,680	0.50	65,603,840	157,680	0.240351784%
314	99,023,040	0.50	65,761,520	157,680	0.239775480%
315	99,338,400	0.50	65,919,200	157,680	0.239201932%
316	99,653,760	0.50	66,076,880	157,680	0.238631122%
317	99,969,120	0.50	66,234,560	157,680	0.238063029%
318	100,284,480	0.50	66,392,240	157,680	0.237497635%
319	100,599,840	0.50	66,549,920	157,680	0.236934920%
320	100,915,200	0.50	66,707,600	157,680	0.236374866%
321	101,230,560	0.50	66,865,280	157,680	0.235817453%
322	101,545,920	0.50	67,022,960	157,680	0.235262662%
323	101,861,280	0.50	67,180,640	157,680	0.234710476%
324	102,176,640	0.50	67,338,320	157,680	0.234160876%
325	102,492,000	0.50	67,496,000	157,680	0.233613844%
326	102,807,360	0.50	67,653,680	157,680	0.233069361%
327	103,122,720	0.50	67,811,360	157,680	0.232527411%
328	103,438,080	0.50	67,969,040	157,680	0.231987976%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
329	103,753,440	0.50	68,126,720	157,680	0.231451037%
330	104,068,800	0.50	68,284,400	157,680	0.230916578%
331	104,384,160	0.50	68,442,080	157,680	0.230384582%
332	104,699,520	0.50	68,599,760	157,680	0.229855032%
333	105,014,880	0.50	68,757,440	157,680	0.229327910%
334	105,330,240	0.50	68,915,120	157,680	0.228803200%
335	105,645,600	0.50	69,072,800	157,680	0.228280886%
336	105,960,960	0.50	69,230,480	157,680	0.227760952%
337	106,276,320	0.50	69,388,160	157,680	0.227243380%
338	106,591,680	0.50	69,545,840	157,680	0.226728155%
339	106,907,040	0.50	69,703,520	157,680	0.226215261%
340	107,222,400	0.50	69,861,200	157,680	0.225704683%
341	107,537,760	0.50	70,018,880	157,680	0.225196404%
342	107,853,120	0.50	70,176,560	157,680	0.224690409%
343	108,168,480	0.50	70,334,240	157,680	0.224186683%
344	108,483,840	0.50	70,491,920	157,680	0.223685211%
345	108,799,200	0.50	70,649,600	157,680	0.223185977%
346	109,114,560	0.50	70,807,280	157,680	0.222688966%
347	109,429,920	0.50	70,964,960	157,680	0.222194165%
348	109,745,280	0.50	71,122,640	157,680	0.221701557%
349	110,060,640	0.50	71,280,320	157,680	0.221211128%
350	110,376,000	0.50	71,438,000	157,680	0.220722865%
351	110,691,360	0.50	71,595,680	157,680	0.220236752%
352	111,006,720	0.50	71,753,360	157,680	0.219752775%
353	111,322,080	0.50	71,911,040	157,680	0.219270921%
354	111,637,440	0.50	72,068,720	157,680	0.218791176%
355	111,952,800	0.50	72,226,400	157,680	0.218313525%
356	112,268,160	0.50	72,384,080	157,680	0.217837956%
357	112,583,520	0.50	72,541,760	157,680	0.217364453%
358	112,898,880	0.50	72,699,440	157,680	0.216893005%
359	113,214,240	0.50	72,857,120	157,680	0.216423597%
360	113,529,600	0.50	73,014,800	157,680	0.215956217%
361	113,844,960	0.50	73,172,480	157,680	0.215490851%
362	114,160,320	0.50	73,330,160	157,680	0.215027487%
363	114,475,680	0.50	73,487,840	157,680	0.214566111%
364	114,791,040	0.50	73,645,520	157,680	0.214106710%
365	115,106,400	0.50	73,803,200	157,680	0.213649273%
366	115,421,760	0.50	73,960,880	157,680	0.213193786%
367	115,737,120	0.50	74,118,560	157,680	0.212740237%
368	116,052,480	0.50	74,276,240	157,680	0.212288613%
369	116,367,840	0.50	74,433,920	157,680	0.211838904%
370	116,683,200	0.50	74,591,600	157,680	0.211391095%
371	116,998,560	0.50	74,749,280	157,680	0.210945176%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
372	117,313,920	0.50	74,906,960	157,680	0.210501134%
373	117,629,280	0.50	75,064,640	157,680	0.210058957%
374	117,944,640	0.50	75,222,320	157,680	0.209618634%
375	118,260,000	0.50	75,380,000	157,680	0.209180154%
376	118,575,360	0.50	75,537,680	157,680	0.208743504%
377	118,890,720	0.50	75,695,360	157,680	0.208308673%
378	119,206,080	0.50	75,853,040	157,680	0.207875650%
379	119,521,440	0.50	76,010,720	157,680	0.207444424%
380	119,836,800	0.50	76,168,400	157,680	0.207014983%
381	120,152,160	0.50	76,326,080	157,680	0.206587316%
382	120,467,520	0.50	76,483,760	157,680	0.206161413%
383	120,782,880	0.50	76,641,440	157,680	0.205737262%
384	121,098,240	0.50	76,799,120	157,680	0.205314853%
385	121,413,600	0.50	76,956,800	157,680	0.204894174%
386	121,728,960	0.50	77,114,480	157,680	0.204475217%
387	122,044,320	0.50	77,272,160	157,680	0.204057969%
388	122,359,680	0.50	77,429,840	157,680	0.203642420%
389	122,675,040	0.50	77,587,520	157,680	0.203228560%
390	122,990,400	0.50	77,745,200	157,680	0.202816380%
391	123,305,760	0.50	77,902,880	157,680	0.202405867%
392	123,621,120	0.50	78,060,560	157,680	0.201997014%
393	123,936,480	0.50	78,218,240	157,680	0.201589808%
394	124,251,840	0.50	78,375,920	157,680	0.201184241%
395	124,567,200	0.50	78,533,600	157,680	0.200780303%
396	124,882,560	0.50	78,691,280	157,680	0.200377983%
397	125,197,920	0.50	78,848,960	157,680	0.199977273%
398	125,513,280	0.50	79,006,640	157,680	0.199578162%
399	125,828,640	0.50	79,164,320	157,680	0.199180641%
400	126,144,000	0.50	79,322,000	157,680	0.198784700%
401	126,459,360	0.50	79,479,680	157,680	0.198390331%
402	126,774,720	0.50	79,637,360	157,680	0.197997523%
403	127,090,080	0.50	79,795,040	157,680	0.197606267%
404	127,405,440	0.50	79,952,720	157,680	0.197216555%
405	127,720,800	0.50	80,110,400	157,680	0.196828377%
406	128,036,160	0.50	80,268,080	157,680	0.196441724%
407	128,351,520	0.50	80,425,760	157,680	0.196056587%
408	128,666,880	0.50	80,583,440	157,680	0.195672957%
409	128,982,240	0.50	80,741,120	157,680	0.195290826%
410	129,297,600	0.50	80,898,800	157,680	0.194910184%
411	129,612,960	0.50	81,056,480	157,680	0.194531023%
412	129,928,320	0.50	81,214,160	157,680	0.194153335%
413	130,243,680	0.50	81,371,840	157,680	0.193777110%
414	130,559,040	0.50	81,529,520	157,680	0.193402341%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
415	130,874,400	0.50	81,687,200	157,680	0.193029018%
416	131,189,760	0.50	81,844,880	157,680	0.192657134%
417	131,505,120	0.50	82,002,560	157,680	0.192286680%
418	131,820,480	0.50	82,160,240	157,680	0.191917648%
419	132,135,840	0.50	82,317,920	157,680	0.191550029%
420	132,451,200	0.50	82,475,600	157,680	0.191183817%
421	132,766,560	0.50	82,633,280	157,680	0.190819002%
422	133,081,920	0.50	82,790,960	157,680	0.190455576%
423	133,397,280	0.50	82,948,640	157,680	0.190093533%
424	133,712,640	0.50	83,106,320	157,680	0.189732863%
425	134,028,000	0.50	83,264,000	157,680	0.189373559%
426	134,343,360	0.50	83,421,680	157,680	0.189015613%
427	134,658,720	0.50	83,579,360	157,680	0.188659018%
428	134,974,080	0.50	83,737,040	157,680	0.188303766%
429	135,289,440	0.50	83,894,720	157,680	0.187949850%
430	135,604,800	0.50	84,052,400	157,680	0.187597261%
431	135,920,160	0.50	84,210,080	157,680	0.187245992%
432	136,235,520	0.50	84,367,760	157,680	0.186896037%
433	136,550,880	0.50	84,525,440	157,680	0.186547387%
434	136,866,240	0.50	84,683,120	157,680	0.186200036%
435	137,181,600	0.50	84,840,800	157,680	0.185853976%
436	137,496,960	0.50	84,998,480	157,680	0.185509200%
437	137,812,320	0.50	85,156,160	157,680	0.185165700%
438	138,127,680	0.50	85,313,840	157,680	0.184823471%
439	138,443,040	0.50	85,471,520	157,680	0.184482504%
440	138,758,400	0.50	85,629,200	157,680	0.184142792%
441	139,073,760	0.50	85,786,880	157,680	0.183804330%
442	139,389,120	0.50	85,944,560	157,680	0.183467109%
443	139,704,480	0.50	86,102,240	157,680	0.183131124%
444	140,019,840	0.50	86,259,920	157,680	0.182796367%
445	140,335,200	0.50	86,417,600	157,680	0.182462832%
446	140,650,560	0.50	86,575,280	157,680	0.182130511%
447	140,965,920	0.50	86,732,960	157,680	0.181799399%
448	141,281,280	0.50	86,890,640	157,680	0.181469489%
449	141,596,640	0.50	87,048,320	157,680	0.181140773%
450	141,912,000	0.50	87,206,000	157,680	0.180813247%
451	142,227,360	0.50	87,363,680	157,680	0.180486903%
452	142,542,720	0.50	87,521,360	157,680	0.180161734%
453	142,858,080	0.50	87,679,040	157,680	0.179837735%
454	143,173,440	0.50	87,836,720	157,680	0.179514900%
455	143,488,800	0.50	87,994,400	157,680	0.179193221%
456	143,804,160	0.50	88,152,080	157,680	0.178872694%
457	144,119,520	0.50	88,309,760	157,680	0.178553311%

Year	End Height	Next Reward	Scheduled Supply	Next-Year Issuance	Annualized Rate
458	144,434,880	0.50	88,467,440	157,680	0.178235066%
459	144,750,240	0.50	88,625,120	157,680	0.177917954%
460	145,065,600	0.50	88,782,800	157,680	0.177601968%
461	145,380,960	0.50	88,940,480	157,680	0.177287103%
462	145,696,320	0.50	89,098,160	157,680	0.176973352%
463	146,011,680	0.50	89,255,840	157,680	0.176660709%
464	146,327,040	0.50	89,413,520	157,680	0.176349170%
465	146,642,400	0.50	89,571,200	157,680	0.176038727%
466	146,957,760	0.50	89,728,880	157,680	0.175729375%
467	147,273,120	0.50	89,886,560	157,680	0.175421109%
468	147,588,480	0.50	90,044,240	157,680	0.175113922%
469	147,903,840	0.50	90,201,920	157,680	0.174807809%
470	148,219,200	0.50	90,359,600	157,680	0.174502765%
471	148,534,560	0.50	90,517,280	157,680	0.174198783%
472	148,849,920	0.50	90,674,960	157,680	0.173895858%
473	149,165,280	0.50	90,832,640	157,680	0.173593986%
474	149,480,640	0.50	90,990,320	157,680	0.173293159%
475	149,796,000	0.50	91,148,000	157,680	0.172993373%
476	150,111,360	0.50	91,305,680	157,680	0.172694623%
477	150,426,720	0.50	91,463,360	157,680	0.172396903%
478	150,742,080	0.50	91,621,040	157,680	0.172100208%
479	151,057,440	0.50	91,778,720	157,680	0.171804532%
480	151,372,800	0.50	91,936,400	157,680	0.171509870%
481	151,688,160	0.50	92,094,080	157,680	0.171216217%
482	152,003,520	0.50	92,251,760	157,680	0.170923568%
483	152,318,880	0.50	92,409,440	157,680	0.170631918%
484	152,634,240	0.50	92,567,120	157,680	0.170341262%
485	152,949,600	0.50	92,724,800	157,680	0.170051594%
486	153,264,960	0.50	92,882,480	157,680	0.169762909%
487	153,580,320	0.50	93,040,160	157,680	0.169475203%
488	153,895,680	0.50	93,197,840	157,680	0.169188470%
489	154,211,040	0.50	93,355,520	157,680	0.168902707%
490	154,526,400	0.50	93,513,200	157,680	0.168617906%
491	154,841,760	0.50	93,670,880	157,680	0.168334065%
492	155,157,120	0.50	93,828,560	157,680	0.168051178%
493	155,472,480	0.50	93,986,240	157,680	0.167769239%
494	155,787,840	0.50	94,143,920	157,680	0.167488246%
495	156,103,200	0.50	94,301,600	157,680	0.167208192%
496	156,418,560	0.50	94,459,280	157,680	0.166929073%
497	156,733,920	0.50	94,616,960	157,680	0.166650884%
498	157,049,280	0.50	94,774,640	157,680	0.166373621%
499	157,364,640	0.50	94,932,320	157,680	0.166097279%
500	157,680,000	0.50	95,090,000	157,680	0.165821853%

Appendix B. Formula Reference

atoms per ATHO = 100,000,000

blocks per day = 864

blocks per nominal year = 315,360

bootstrap supply =

$$1,250,000 \times (8 + 4 + 2 + 1)$$
$$= 18,750,000 \text{ ATHO}$$

tail annual issuance =

$$315,360 \times 0.50$$
$$= 157,680 \text{ ATHO}$$

year-500 expected blocks =

$$500 \times 315,360$$
$$= 157,680,000$$

year-500 supply =

$$18,750,000$$
$$+ (157,680,000 - 5,000,000) \times 0.50$$
$$= 95,090,000 \text{ ATHO}$$

required transaction fee =

$$\max(600 \text{ atoms}, \text{full serialized bytes} \times 1 \text{ atom})$$

miner claim limit =

$$\text{active subsidy} + \text{validated included fees}$$

Appendix C. Glossary

- **annualized issuance:** reward at the modeled height multiplied by the expected 315,360 blocks per nominal year
- **annualized issuance rate:** next-year annualized issuance divided by current scheduled supply
- **atom:** smallest Atho accounting unit, equal to 0.00000001 ATHO
- **bootstrap issuance:** subsidy from the four finite 8, 4, 2, and 1 ATHO eras
- **burn:** provable removal of spendability; current Atho fees are not burned
- **coinbase:** first transaction in a block, which claims allowed subsidy and included fees
- **fee floor:** minimum fee required for a non-coinbase transaction
- **optional tip:** fee paid above the required floor
- **scheduled supply:** cumulative block subsidy permitted through a height
- **tail emission:** permanent 0.50 ATHO subsidy after the bootstrap eras
- **TX-PoW:** transaction-level anti-spam proof of work, separate from fees and block proof of work